Leveraging Deep Learning and Blockchain for Enhanced Transparency and Traceability in the Indian Herbal Product Supply Chain

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***Abstract*—Medicinal plants have been a cornerstone of healthcare practices globally, serving a significant population in developing countries who rely on affordable herbal remedies. However, the current marketing system through Mandis and wholesale markets suffers from opacity. Numerous intermediaries create an information gap for farmers regarding fair pricing and market trends. Additionally, concerns exist about misidentification and adulteration of medicinal plants due to limited knowledge among collectors and traders. The National Medicinal Plants Board (NMPB) recognizes the need for a robust supply chain. This paper proposes a blockchain-based platform 'Indian Herbal Blockchain Network (IHBN)' integrated with Deep Learning for image processing. The deep learning model has achieved a decent accuracy of 90.24% which decreases the chance of misidentification. The platform ensures transparent traceability of medicinal plants from source to consumer, along with accurate identification using image analysis. This system promotes transparency within the medicinal plant supply chain, reducing the risk of adulteration and improving overall trust.**

Keywords*—*blockchain, deep learning, medicinal plant supply chain, traceability, image processing

I. Introduction

The global demand for medicinal plants has increased significantly in recent years. With over 8,000 species of medicinal plants, India's medicinal plant industry boasts a market size of $9 Bn. India has even exported Ayush and Herbal products worth $1,240.6 Mn in the last two years (2021-2022 to 2022-23). However, concerns regarding the authenticity and quality of these products are also on the rise. Study [15] talks in detail about the present-day prevailing trend of adulteration and substitution in Indian medicinal plants. This means consumers may unknowingly purchase products containing inferior or even harmful ingredients due to adulteration. These issues arise due to a lack of transparency in the supply chain, making it difficult to track the origin and processing of medicinal plants. This lack of transparency in the supply chain can have serious health implications for consumers.

Blockchain technology offers a promising solution to address these concerns. Blockchain is a secure, distributed ledger system that creates an immutable record of transactions. Study [16] describes the process of designing a supply chain management system using Blockchain. In the context of the medicinal plant supply chain, each participant – plantation base, trader, manufacturer, and retailer – acts as a node on the network. Every step a plant takes on its journey, from harvest to sale, is recorded as a block on the blockchain. This shared ledger ensures transparency and eliminates the possibility of tampering with data. Each stakeholder can access relevant information about the plant, fostering trust and collaboration throughout the supply chain.

This paper proposes a novel Blockchain-based traceability system - Indian Herbal Blockchain Network (IHBN), specifically designed for the medicinal plant supply chain in India. The system leverages Deep Learning for image recognition to enhance the accuracy of plant identification. This paper details the system architecture, functionalities, and stakeholders involved. We have used 20 classes of Medicinal Leaf Dataset for training. Additionally, the paper discusses the potential benefits and limitations of the proposed system, concluding with future development directions.

II. Related Work

In [1] HerBChain is introduced as a blockchain-based platform designed to improve the quality control and authenticity of herbal products by securely recording every manufacturing and supply detail. Utilizing a consortium blockchain model and adhering to various international standards, HerBChain aims to ensure product traceability and minimize manipulation throughout the supply chain. Despite its potential, the platform faces limitations such as limited technological accessibility and familiarity among stakeholders, resistance to adoption within traditional industry segments, challenges in achieving uniform compliance across different regions, and the high costs associated with initial setup and ongoing maintenance. These factors might affect the platform's widespread implementation and effectiveness in quality control.

Study [2] investigates the role of smart contracts in fortifying the pharmaceutical supply chain in Belarus, a pioneer in legalizing such contracts. It details how smart contracts can mitigate prevalent issues like documentation inaccuracies, quality defects, and temperature control failures by automating transactions and ensuring transparency. However, it faces implementation challenges including user resistance, bureaucratic barriers, significant infrastructural costs, and the need for extensive network participation. The study underlines the necessity of overcoming these hurdles to leverage smart contracts' full potential in enhancing the supply chain's integrity and efficiency.

The research paper [3] proposes an automated blockchain and biometric-based system to enhance the integrity and distribution of India's Mid-Day Meal Scheme, addressing the prevalent issues of fraud, manual errors, and opacity. Utilizing

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Ethereum smart contracts and fingerprint identification, alongside QR codes for tracking, it aims to secure and transparently record transactions, thereby reducing corruption and ensuring meals reach the correct recipients. However, challenges such as limited access to required technologies in rural areas, the need for widespread blockchain comprehension, substantial initial setup costs, and potential data privacy concerns could hinder the system's implementation and effectiveness.

Study [4] introduces a blockchain-based supervisory model aimed at combating herbal medicine counterfeiting. By ensuring the integrity of herbal medicine's production, processing, and distribution stages through immutability and traceability, along with segmented encryption for data protection, it seeks to reduce counterfeiting while preserving product quality by monitoring environmental factors. However, the model's widespread adoption is challenged by the need for substantial technological infrastructure, the complexity of blockchain technology and its acceptance, and concerns over data privacy and security.

Study [5] introduces a double-blockchain traceability system for enhancing the supply chain transparency of rare Chinese herbal medicines, leveraging a private blockchain for data collection merged with a consortium blockchain for data verification and tamper prevention. Information is stored securely in the Interplanetary File System (IPFS), improving supply chain and providing comprehensive traceability to consumers. However, the system's adoption faces challenges such as the complexity of blockchain technology, high initial and maintenance costs, data privacy concerns, and potential resistance from stakeholders unready to alter existing systems.

Research paper [6] presents the creation of an intelligent application using DenseNet-121 for recognizing 61 types of Chinese herbal medicines with over 94% accuracy, aiming to improve identification efficiency and merge traditional with modern medicine knowledge. While it showcases substantial advancements in utilizing deep learning for herbal medicine identification, the system faces particular limitations such as: variability in herbal sample appearance not fully represented in the dataset, possible inadequacy of data augmentation in covering real-world variations, limited generalization to unseen herbal types, and a strong dependency on the DenseNet architecture, which may not uniformly suit all recognition tasks.

Study [7] presents an extensive examination of methodologies ranging from clustering and colour imaging analysis to sophisticated machine learning techniques like ANN, SVM, DNN, and CNN for plant disease detection, showcasing their promise in enhancing plant health diagnostics through high accuracy rates. The study recognizes major challenges like the hard task of making these models work for all plant types, their high setup costs and complexity, their need for a lot of varied data, and the necessity to constantly update them to fight new plant diseases. This highlights the importance of continued work to make these methods more accessible and useful, especially where resources are limited.

Research paper [8] discusses the limitations of traditional methods in diagnosing plant diseases and presents a digital approach involving image pre-processing, segmentation, feature extraction, and classification using machine learning models like CNN, KNN, and logistic regression. This process aims to improve the accuracy and efficiency of disease prediction. However, it faces challenges such as dependence on large, diverse datasets, difficulty in generalizing findings to all plant species and conditions, the necessity for significant computational resources and expertise, and the need for ongoing model updates due to the evolving nature of plant diseases. These limitations underscore the balance between innovation and practical application constraints.

Study [9] introduces an efficient deep learning methodology utilizing Convolutional Neural Networks (CNN) and Logistic Decision Regression (LDR) for automated diagnosis of plant diseases from images. This novel approach targets to overcome the inefficiencies and high resource requirements of traditional methods through automated monitoring and image analysis, aiming for enhanced accuracy and specificity in disease detection. However, it faces challenges including data limitations from sources like Kaggle, difficulties in generalizing across diverse plant species, significant computational demands for CNN training, and the need for continuous updates to address evolving plant diseases.

The methodology detailed in study [10] incorporates a sequential approach starting with image preprocessing and culminating in classification through machine learning (CNN and ANN), achieving a notable accuracy of 93.02% on a custom dataset. The process emphasizes noise reduction, segmentation, and feature normalization, preparing images for efficient analysis. However, the technique faces limitations in handling complex scenarios such as real-time imagery or clustered leaves, which could impact accuracy due to challenges in segmentation and feature extraction. This highlights the need for technique refinement and advanced methods to extend the model's applicability to a broader array of image contexts, indicating a significant area for future research.

The research paper [11] presents a methodology for identifying plant species that combines Internet of Things (IoT) sensors for data collection with a Deep-Convolutional Neural Network (D-CNN), using region splitting and merging to enhance accuracy by focusing on unique plant features. While innovative, this approach faces challenges such as high costs of IoT hardware, sensitivity to data quality and environmental factors, substantial computational needs, D-CNN model complexity requiring expertise, limitations in identifying unseen species, and the necessity for regular updates for new species, affecting scalability and practical use in resource-constrained environments.

Study [12] assesses how different image filtering techniques affect an Xception CNN model's ability to identify herbal plants. Color Saturation emerged as the most effective technique, achieving perfect accuracy. However, the study notes limitations such as issues with generalizing across various plant species and conditions, the demanding computational resources for processing and training, and the reliance on high-quality images. This research highlights the balance between improved accuracy through technological advancements and the practical challenges of applying and scaling these methods in diverse environments.

Research Paper [13] presents a methodology combining Canny Edge Detection and machine learning classifiers (SVM and KNN) for classifying medicinal leaves through processing leaf image features like shape and size. Challenges include

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potential classification limitations for diverse leaf shapes, the need for high-quality images, and the high computational demand for processing large datasets. Additionally, the choice between SVM and KNN classifiers could impact classification accuracy, questioning adaptability to various species and conditions.

Study [14] examines herbal leaf classification through a comparative analysis using object detection algorithms (RCNN, Fast RCNN, Faster RCNN) and deep learning models (CNN, RNN, LSTM, BiLSTM). The study involved collecting images of four herbal leaf classes, applying these algorithms for feature extraction and classification. While showcasing a comprehensive approach to improve identification accuracy and speed, it is limited by its focus on a select few models, omitting a wider comparison with other modern methods, which could impact the evaluation of its effectiveness and scalability in various scenarios.

III. Proposed System

The proposed solution is to enhance transparency and trust in the supply chain of herbal and medicinal products, where the power of blockchain is harnessed to ensure that the data isn't tampered with and is genuine.

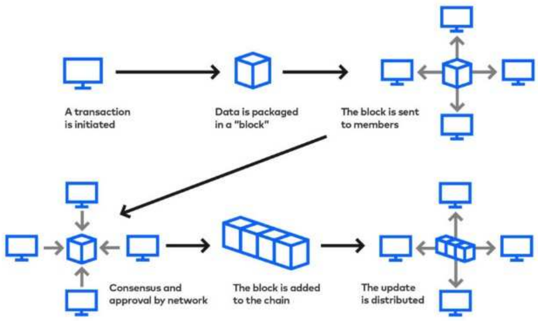

Fig. 1. Blockchain Architecture.

A. Selection of Stakeholders

Admission criteria are employed to select potential users to join the platform. The potential user is assessed by using a set of measurable standards already set up by the National Medicinal Plants Board (NMPB) and the Quality Council of India (QCI) as per the Voluntary Certification Scheme for Medicinal Plant Produce (VCSMPP). This certification encourages Good Agricultural Practices (GAP) and Good Field Collection Practices (GFCP) in medicinal plants and enhances the quality and safety of these plants. In the platform, there are four nodes/stakeholders:

- *Plantation Base:* Registers with the system and provides details about the medicinal plant including corresponding image data.
- *Trader:* Purchases the harvest from farmers, scans a QR code linked to the harvest record and verifies the produce via in-house ML model.
- *Manufacturer:* Receives the medicinal plants from traders, scans a QR code linked to the blockchain record for verification, and logs processing details.
- *Retailer:* Receives the medicinal plants from traders, scans a QR code linked to the blockchain record.

B. Workflow of Indian Herbal Blockchain Network

IHBN facilitates the record keeping of a Herbal Product from farm to store. The system assigns accounts to authorized personnel within admitted stakeholder organizations. For example, if a company participates in the "Plantation Base" node, their executive staff can add new records for herbal material plantations. Upon record submission, the system generates a unique identifier for each entry. The system automatically generates a unique QR code linked to the specific record. This QR code can be downloaded and printed for inclusion on the product packaging, allowing consumers / stakeholders to access detailed information.

Thus, a new QR code is generated at each node, which has all the information linking back to the first node. Data entry by the Retailer node leads to the final QR code generation, which is accessible to the customers buying the product. The workflow can be elaborated in the following manner:

1) Harvest Recording: Staff in the Plantation Base node will upload information about the harvested medicinal plant. The system generates a unique identifier and stores the record on the blockchain.

2) Sale to Trader: Plantation Base sells the harvest to a registered trader.

3) Trader Verification: Upon receiving the harvest, the trader uses IHBN's user interface to upload pictures of the received plants. IHBN's in-house deep learning model analyzes the pictures and verifies if the received harvest matches the species recorded in the Plantation Base node.

a) Match: If the analysis confirms the plant, the process continues. The staff in the Trader node uploads information at this stage to the blockchain.

b) Mismatch: The system flags a potential misidentification. The trader can contact the Plantation Base for clarification or reject the harvest.

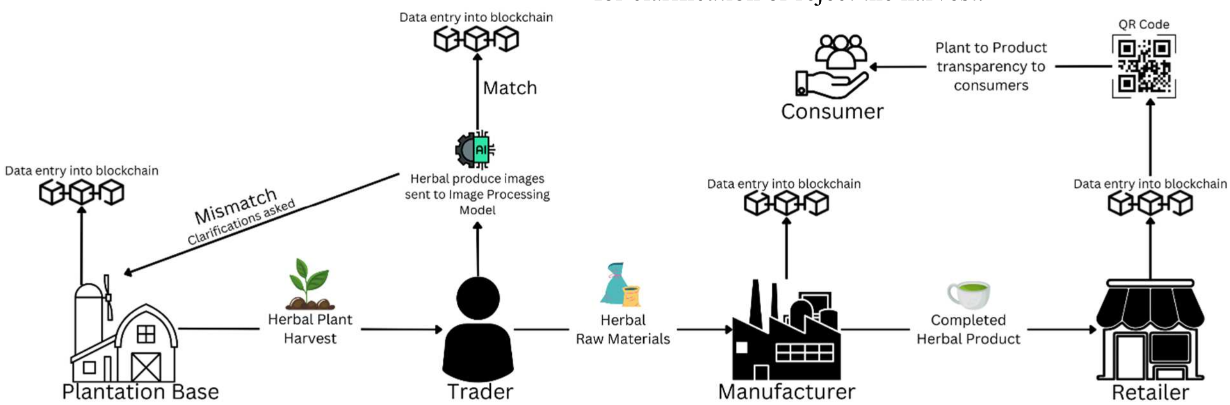

Fig. 2. Proposed Solution - Architecture Diagram.

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4) Sale to Manufacturer: Trader sells the verified medicinal plants to a registered manufacturer.

5) Manufacturing Process: The manufacturer logs the processing details for the medicinal plants. This information is linked to the existing records on the blockchain.

6) Sale to Retailer: The manufacturer sells the finished product (e.g., herbal tea) to a registered retailer.

7) Retailer and Consumer: The retailer adds the final node specific information and a final QR code is generated.

The consumer can access product information and history by scanning this QR code on the packaging, which links back to the initial harvest record on the blockchain.

C. Implementation Details

The backend of IHBN leverages Hyperledger Fabric, an open-source platform from the Linux Foundation designed for secure and scalable blockchain solutions. This modular architecture offers high confidentiality, flexibility, and resilience. Hyperledger Fabric operates as a network of nodes, each performing specific tasks like transaction validation, ledger maintenance, and smart contract execution (called "Chaincode"). A consensus mechanism ensures data integrity and consistency through transaction validation and ordering. Study [17] provided guidance to practically implement Hyperledger Fabric. HerbServer, built with NodeJS and Express.js, bridges the user interface and the blockchain network. Functioning as a web server, it interacts with Hyperledger Fabric via the Fabric-Client SDK. Additionally, HerbServer exposes a REST API (GET and POST endpoints) that simplifies interaction with the blockchain network. Study [18] was taken as reference when developing the backend.

The system utilizes a progressive web app (PWA) developed using React.js as its user interface layer. PWAs provide multiple advantages over traditional web applications as discussed in [19]. PWAs leverage web platform technologies, offering a user experience comparable to native apps across various devices (phones, tablets, desktops). IHBN's PWA provides a user-friendly interface catering to diverse stakeholders within the supply chain. Key functionalities include:

- *Data Entry for Authorized Users:* Each node (e.g., Plantation Base) within the network can securely enter relevant data through the PWA interface.
- *QR Code Scanning for Verification:* Consumers can easily verify product authenticity using the integrated QR code scanner.
- *Offline Functionality:* The PWA offers offline functionalities, allowing users to access previously scanned product information even without an internet connection.

Another important component is an image recognition model designed to efficiently identify medicinal plants. To achieve this, a diverse set of high-quality plant images of 20 classes (Aloe vera, Ashoka, Amla, Castor etc.) are collected, focusing on capturing a wide variety. These images undergo a standardization process that includes resizing, augmentation, and colour normalization. The study [20] was taken to build an approach on leveraging Convolutional Neural Network (CNN) based deep learning technique to attain efficiency in classifying medicinal leaves. A custom CNN with Sequential architecture is then employed for feature extraction due to its accuracy and efficiency. This streamlined process empowers the deep learning model to effectively authenticate plant data against Blockchain records. This significantly enhances the system's integrity and reliability in verifying plant species information provided by traders.

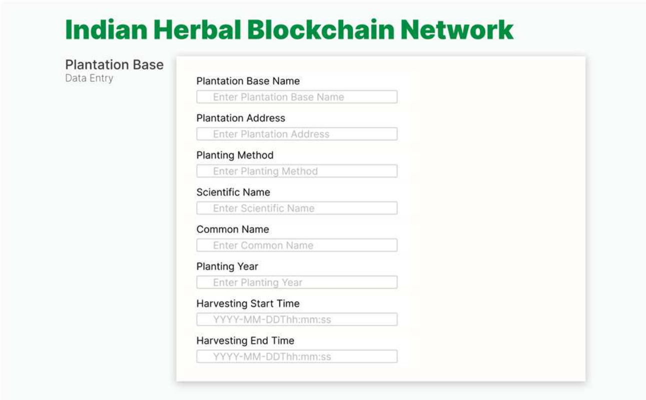

Fig. 3. Data Entry in the Plantation Base Node.

Fig. 4. Product Data Accessible via QR Code.

Fig. 5. Harvest Verification via in-house Deep Learning Model.

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TABLE I. CURRENT SYSTEM VS. IHBN

Feature	Current System	IHBN
Marketing/Supply Chain	Opaque, Multi-layered (Mandis, etc.)	Transparent, Blockchain-based
Data Accessibility	Limited, Price Data Not Readily Available	Real-time Price Information accessible to stakeholders
Due Diligence	Paper-based Records, Annual Reporting	Secure, Immutable Data on Blockchain
Data Reliability	Potential for Tampering	Tamper-proof Data Records
Plant Identification	Manual Verification, Potential for Errors	Deep Learning Model for Accurate Identification

IV. Results And Discussion

The growth of the Herbal Product Industry has underscored the critical need for transparency and trust in the supply chain. Insights from previous studies [1] and [2] have inspired the development of a tailored solution for the Indian herbal product industry, known as the Indian Herbal Blockchain Network (IHBN). IHBN has demonstrated high levels of data accuracy and satisfactory transaction speed, as evidenced by positive user feedback.

Our system, IHBN, builds upon prior work like HerBChain [1] by incorporating a deep learning image classification model. This additional layer enhances the overall reliability of the system by verifying the authenticity of medicinal plants at the point of harvest. We have achieved the accuracy of 90.24% and loss of 0.3161 on the validation set of the data.

V. Conclusion

In conclusion, the Indian Herbal Blockchain Network (IHBN) leverages Hyperledger Fabric for a secure and transparent medicinal plant supply chain. By integrating NodeJS, React.js, and a sophisticated image recognition model based on Sequential CNN, IHBN ensures reliable plant identification and seamless interaction with the blockchain, fostering trust and authenticity in the industry.

VI. Future Work

The future development of the Indian Herbal Blockchain Network (IHBN) will focus on crucial enhancements including scalability and performance optimization to manage increased transactions, the incorporation of advanced machine learning algorithms for better accuracy in plant identification and health diagnosis, and the integration of IoT devices for seamless data collection. Additionally, efforts will be directed towards expanding the ecosystem to include a wider range of stakeholders such as certification authorities and consumers, fostering cross-chain interoperability for broader applications, and conducting comprehensive studies on the societal and environmental impacts. It's crucial to track the accuracy and loss scores of our deep learning model. Currently, we're achieving an accuracy of 0.9024 with a loss of 0.3161, which shows strong performance. However, continuous monitoring and potential improvements are essential for optimal results. These targeted improvements aim to elevate the platform's capabilities and ensure its positive contribution to the herbal market's transparency, security, and efficiency.

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